

# XERIN MARKET

*Africa's Premier E-Commerce Marketplace*

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## SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

|                |              |
|----------------|--------------|
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*Connecting Buyers, Sellers & Xerin Logistics Across Africa*

## Document Revision History

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| 1.0.0   | June 2026 | Initial Release — Full Requirements Document | Xerin Product Team |

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# 1. Introduction & Overview

## 1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional, technical, and non-functional requirements for Xerin Market — a pan-African multi-vendor e-commerce marketplace. It serves as the authoritative reference for all design, development, testing, and delivery teams involved in building the platform.

## 1.2 Project Vision

Xerin Market is Africa's premier multi-vendor e-commerce marketplace designed to operate on a model similar to Amazon. The platform aims to connect millions of buyers and sellers across Africa on a single, trusted digital commerce infrastructure — powered by Xerin Logistics for seamless fulfilment.

## 1.3 Scope

The Xerin Market platform encompasses:

- A buyer-facing storefront with product discovery, cart, and checkout
- A seller portal for product listing, order management, and analytics
- An administrator panel for platform governance and reporting
- Integration with African mobile money networks (M-Pesa, Airtel Money, Tigo Pesa, etc.)
- Integration with Xerin Logistics for real-time shipment tracking
- Multi-country and multi-currency support across African markets

## 1.4 Definitions & Acronyms

| Term / Acronym | Definition                                                     |
|----------------|----------------------------------------------------------------|
| SRS            | Software Requirements Specification                            |
| MVP            | Minimum Viable Product                                         |
| API            | Application Programming Interface                              |
| JWT            | JSON Web Token — used for stateless authentication             |
| OAuth          | Open Authorization — third-party login protocol                |
| SKU            | Stock Keeping Unit — unique product identifier                 |
| POS            | Point of Sale                                                  |
| KYC            | Know Your Customer — seller verification process               |
| USSD           | Unstructured Supplementary Service Data — mobile money channel |

|            |                           |
|------------|---------------------------|
| <b>CDN</b> | Content Delivery Network  |
| <b>SLA</b> | Service Level Agreement   |
| <b>2FA</b> | Two-Factor Authentication |

## 2. System Overview & Architecture

### 2.1 Platform Description

Xerin Market is a web-first, mobile-responsive marketplace platform delivered as a Progressive Web Application (PWA). The system follows a microservices-inspired architecture to ensure scalability, independent deployability, and fault isolation.

### 2.2 High-Level Architecture

| Layer          | Components                                                                        |
|----------------|-----------------------------------------------------------------------------------|
| Presentation   | React / Next.js — SSR & SSG for fast page loads; Tailwind CSS; Mobile-first PWA   |
| API Gateway    | Express.js REST API with rate limiting, authentication middleware, and versioning |
| Business Logic | Microservices: Auth, Product, Order, Payment, Logistics, Notification, Analytics  |
| Data Storage   | PostgreSQL (relational data); Redis (caching & sessions); Elasticsearch (search)  |
| File Storage   | AWS S3 or equivalent for product images, documents, and media assets              |
| Messaging      | Message queue (e.g., RabbitMQ / AWS SQS) for async notifications and events       |
| CDN            | CloudFront or equivalent for static assets and image optimisation                 |

### 2.3 Deployment Model

- Cloud-hosted infrastructure (AWS / GCP / Azure)
- Containerised services via Docker and Kubernetes for orchestration
- CI/CD pipeline with automated testing and blue-green deployment
- Multi-region hosting to optimise latency across African markets

### 3. Stakeholders & User Roles

| Role              | Description                                          | Key Permissions                                         |
|-------------------|------------------------------------------------------|---------------------------------------------------------|
| Buyer (Customer)  | Registered end-user browsing and purchasing products | Browse, search, cart, checkout, order tracking, reviews |
| Guest User        | Unregistered visitor on the platform                 | Browse products, view listings, add to cart (limited)   |
| Seller (Vendor)   | Business or individual listing products for sale     | Product CRUD, order management, analytics, payouts      |
| Platform Admin    | Xerin internal administrator                         | Full platform control, user management, reports         |
| Super Admin       | Technical lead with system-level access              | All admin capabilities + system configuration           |
| Logistics Partner | Xerin Logistics team / third-party courier           | View assigned shipments, update delivery status         |
| Support Agent     | Customer service representative                      | View orders, process refunds, manage disputes           |

## 4. Functional Requirements

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### 4.1 User Account Management

#### 4.1.1 Buyer Registration & Login

- Register with email/phone and password
- OAuth login (Google, Facebook, Apple)
- Email and SMS OTP verification
- Password reset via email/SMS
- Profile management: name, address book, profile photo
- Two-Factor Authentication (2FA) option

#### 4.1.2 Seller Registration & Verification

- Separate seller registration flow
- Business name, category, and contact details
- KYC document upload (national ID, business certificate)
- Bank account / mobile money details for payouts
- Seller agreement acceptance
- Admin review and approval workflow

#### 4.1.3 Profile Management

- Update personal / business information
- Address book management (multiple delivery addresses)
- Notification preferences
- Account security settings (password, 2FA)
- Account deactivation and deletion

### 4.2 Product Management

#### 4.2.1 Product Listing

- Add products with title, description, category, and tags
- Upload up to 10 product images per listing (min 800×800px)
- Set base price, discounted price, and currency
- Define stock quantity and low-stock alert threshold
- Product variants: size, colour, material, weight
- SKU auto-generation and manual override
- Bulk product upload via CSV / Excel template

#### 4.2.2 Category & Taxonomy



- Hierarchical category tree (Category > Subcategory > Sub-subcategory)
- Seller assigns product to relevant category
- Admin manages and curates master category list
- Brand registration and association

#### **4.2.3 Product Moderation**

- New listings queued for admin review before publishing
- Automated scan for prohibited or counterfeit items
- Admin approval, rejection with reason, or request for edit
- Seller receives notification on listing status

### **4.3 Search & Discovery**

- Full-text search powered by Elasticsearch
- Auto-complete / search suggestions as user types
- Filter by: price range, category, location, rating, brand, seller
- Sort by: relevance, price ascending/descending, newest, best-selling
- Faceted search with dynamic filter counts
- AI-powered product recommendations (similar items, 'customers also bought')
- Recently viewed products
- Trending and featured product sections on homepage
- Flash sale and promotional banners

### **4.4 Shopping Cart**

- Add to cart from product page and listing pages
- Update item quantity in cart
- Remove individual items or clear cart
- Save items for later (wish list)
- Persistent cart across sessions for logged-in users
- Cart summary with subtotal, estimated shipping, and taxes
- Apply coupon / promo codes
- Cross-sell recommendations in cart sidebar

### **4.5 Checkout & Payments**

#### **4.5.1 Checkout Flow**

1. Review cart items
2. Select / add delivery address
3. Choose delivery method (standard / express / pickup)
4. Select payment method

5. Review order summary
6. Place order and receive confirmation

### 4.5.2 Supported Payment Methods

| Payment Type       | Providers                                 | Markets               |
|--------------------|-------------------------------------------|-----------------------|
| Mobile Money       | M-Pesa, Airtel Money, Tigo Pesa, MTN MoMo | East & West Africa    |
| Card Payments      | Visa, Mastercard, Verve                   | Pan-Africa            |
| USSD Payments      | Carrier-agnostic USSD gateway             | Feature phone markets |
| Bank Transfer      | Direct bank API integration               | All markets           |
| Xerin Pay (Future) | Platform native wallet                    | Pan-Africa — Phase 2  |
| Buy Now Pay Later  | Third-party BNPL partner                  | Selected markets      |

### 4.5.3 Payment Processing

- PCI-DSS compliant payment handling
- Multi-currency support with real-time FX rates
- Automatic payment splitting between seller and platform (commission)
- Order confirmation email and SMS upon successful payment
- Failed payment retry mechanism and user notification

## 4.6 Order Management

### 4.6.1 Buyer Order Management

- Order history with status timeline
- Real-time order tracking integrated with Xerin Logistics
- Order cancellation within defined cancellation window
- Return and refund request initiation
- Download invoice / receipt as PDF

### 4.6.2 Seller Order Management

- View incoming orders with buyer details and items
- Accept, reject, or flag orders for review
- Print packing slip / shipping label
- Mark order as shipped and input tracking number
- Handle return requests and issue refunds

### 4.6.3 Order Status Lifecycle

| Status            | Description                                |
|-------------------|--------------------------------------------|
| Pending Payment   | Order placed but payment not yet confirmed |
| Payment Confirmed | Payment successfully processed             |
| Processing        | Seller is preparing the order for dispatch |
| Dispatched        | Order handed over to logistics partner     |
| In Transit        | Order moving through logistics network     |
| Out for Delivery  | Last-mile delivery in progress             |
| Delivered         | Order received by buyer                    |
| Return Requested  | Buyer initiated return                     |
| Refunded          | Refund issued to buyer                     |
| Cancelled         | Order cancelled by buyer or seller         |

## 4.7 Seller Dashboard

- Overview metrics: total sales, revenue, active listings, pending orders
- Sales performance charts (daily, weekly, monthly, custom range)
- Product performance: views, add-to-cart rate, conversion rate
- Inventory management with restock alerts
- Revenue reports exportable to CSV / PDF
- Payout schedule and transaction history
- Commission and fee breakdown per order
- Seller performance score and badges (Top Seller, Verified, etc.)
- In-platform messaging with buyers

## 4.8 Logistics Integration (Xerin Logistics)

- Automated shipment creation upon order dispatch
- Real-time tracking updates pushed to buyer and seller
- Shipping cost calculator based on origin, destination, weight, and dimensions
- Support for multiple delivery options: standard, express, same-day (where available)
- Pickup point selection for buyers
- Proof of delivery capture (photo / signature)
- Failed delivery and re-delivery scheduling
- Integration via RESTful API with webhook events for status updates
- Third-party courier fallback for markets where Xerin Logistics is not yet operational

## 4.9 Reviews & Ratings

- Buyers can review and rate products (1–5 stars + written review)
- Buyers can rate sellers separately on: communication, packing, speed
- Reviews restricted to verified purchasers (confirmed delivery)
- Sellers can respond publicly to reviews
- Helpful / not helpful voting on reviews
- Automated flagging of suspicious or spam reviews
- Admin moderation queue for reported reviews
- Aggregate star rating prominently displayed on product and seller pages

## 4.10 Notifications

| Notification Event          | Email | SMS | Push |
|-----------------------------|-------|-----|------|
| Order placed / confirmed    | ✓     | ✓   | ✓    |
| Payment successful / failed | ✓     | ✓   | ✓    |
| Order dispatched            | ✓     | ✓   | ✓    |
| Delivery status update      | ✓     | ✓   | ✓    |
| Order delivered             | ✓     | ✓   | ✓    |
| Return / refund update      | ✓     | ✓   | —    |
| New product review received | ✓     | —   | ✓    |
| Low stock alert (seller)    | ✓     | ✓   | ✓    |
| New order received (seller) | ✓     | ✓   | ✓    |
| Promotional offers / deals  | ✓     | —   | ✓    |
| Account security alerts     | ✓     | ✓   | ✓    |

## 4.11 Admin Panel

### 4.11.1 User Management

- View, search, and filter all buyers and sellers
- Suspend, ban, or reinstate user accounts
- Review KYC documents and approve/reject seller applications
- Assign and manage admin roles and permissions

### 4.11.2 Product & Catalogue Management

- Review and approve/reject new product listings
- Edit or remove non-compliant listings

- Manage category tree, brands, and attributes
- Configure flash sales and featured product slots

#### **4.11.3 Order & Dispute Management**

- View all platform orders with full audit trail
- Intervene in buyer-seller disputes
- Manually trigger refunds or order cancellations
- Export order data for reconciliation

#### **4.11.4 Financial Management**

- View platform revenue, commissions, and payouts
- Approve or hold seller payout batches
- Configure commission rates per category or seller tier
- Generate financial reports (daily, monthly, annual)

#### **4.11.5 Analytics Dashboard**

- Platform GMV (Gross Merchandise Value) over time
- Active users, new registrations, and churn metrics
- Top-selling products and categories
- Country and region performance heatmaps
- Funnel analytics: visits → product views → add-to-cart → purchase

## 5. Non-Functional Requirements

| Category      | Requirement         | Target / Detail                                                                       |
|---------------|---------------------|---------------------------------------------------------------------------------------|
| Performance   | Page Load Time      | < 3 seconds on 3G connection (Africa-optimised)                                       |
| Performance   | API Response Time   | < 500ms for 95th percentile under normal load                                         |
| Scalability   | Concurrent Users    | Support 100,000+ concurrent users at launch, horizontally scalable                    |
| Availability  | Uptime SLA          | 99.9% uptime (< 8.7 hours downtime per year)                                          |
| Availability  | Planned Maintenance | Zero-downtime deployments; maintenance during off-peak hours                          |
| Security      | Data Encryption     | TLS 1.3 in transit; AES-256 at rest                                                   |
| Security      | Authentication      | JWT + OAuth 2.0; bcrypt password hashing; 2FA support                                 |
| Security      | Payment Security    | PCI-DSS Level 1 compliance for all payment flows                                      |
| Accessibility | WCAG Compliance     | WCAG 2.1 AA — screen reader support, keyboard navigation                              |
| Localisation  | Languages           | English, Swahili, French, Hausa (Phase 1); expand in Phase 2                          |
| Localisation  | Currencies          | KES, TZS, UGX, NGN, GHS, ZAR, USD + real-time FX                                      |
| SEO           | Search Engine       | Server-side rendering (SSR) for all product and category pages                        |
| Reliability   | Data Backup         | Automated daily backups with 30-day retention and point-in-time recovery              |
| Monitoring    | Observability       | Real-time error tracking, performance monitoring, and alerting (e.g., Datadog/Sentry) |

## 6. Technology Stack

| Layer              | Technology                | Rationale                                                         |
|--------------------|---------------------------|-------------------------------------------------------------------|
| Frontend Framework | React + Next.js 14        | SSR, SSG, ISR for SEO and performance; large ecosystem            |
| Styling            | Tailwind CSS              | Utility-first; rapid UI development; consistent design tokens     |
| State Management   | Zustand / Redux Toolkit   | Lightweight state management for cart, auth, and UI state         |
| Backend Runtime    | Node.js (LTS)             | Non-blocking I/O; shared language with frontend team              |
| API Framework      | Express.js                | Lightweight, mature REST framework with rich middleware ecosystem |
| Primary Database   | PostgreSQL 15+            | ACID compliance; complex relational queries; proven at scale      |
| Search Engine      | Elasticsearch 8           | Full-text search, faceting, and auto-complete at high volume      |
| Caching Layer      | Redis 7                   | Session storage, rate limiting, real-time leaderboards, pub/sub   |
| File Storage       | AWS S3 / Cloudflare R2    | Scalable object storage with CDN integration for images           |
| Authentication     | JWT + OAuth 2.0           | Stateless auth tokens; social login support                       |
| Payment Gateway    | Flutterwave / Paystack    | Pan-African coverage; mobile money + card; proven reliability     |
| SMS Gateway        | Africa's Talking / Twilio | Low-cost SMS across 40+ African countries                         |
| Push Notifications | Firebase Cloud Messaging  | Cross-platform web push for PWA and mobile                        |
| Message Queue      | AWS SQS / RabbitMQ        | Async processing for notifications, emails, and logistics events  |
| DevOps             | Docker + Kubernetes       | Container orchestration; auto-scaling; rolling deployments        |
| CI/CD              | GitHub Actions            | Automated testing, builds, and deployments on every PR            |
| Monitoring         | Datadog / Sentry          | APM, error tracking, uptime alerts, and dashboards                |

## 7. UI/UX Requirements

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### 7.1 Design Principles

- Mobile-first design — Africa's primary internet access is via mobile devices
- Low-bandwidth optimisation — progressive image loading, lazy loading, and offline-capable PWA
- Culturally inclusive design — imagery, colour, and language that resonate across African markets
- Minimal cognitive load — clear CTAs, simple navigation, and guided checkout flows
- Dark mode and light mode with user preference persistence

### 7.2 Design System

- Consistent component library built on Tailwind CSS and shadcn/ui
- Typography: Primary font — Inter or Poppins; Secondary — system fonts as fallback
- Colour palette: Primary Navy, African Gold accent, functional greens and reds for status
- Iconography: Consistent SVG icon set (Lucide or custom Xerin icons)
- Spacing and grid system following 8pt grid principle

### 7.3 Key User Flows

#### Buyer: Discover & Purchase

7. Land on homepage → see featured products and promotions
8. Search or browse by category
9. View product detail page (images, description, reviews, seller info)
10. Add to cart → proceed to checkout
11. Enter address → select delivery → select payment
12. Confirm order → receive email and SMS confirmation

#### Seller: List & Manage

13. Register and complete KYC
14. Activate seller account after admin approval
15. Access seller dashboard → add first product
16. Set prices, upload images, define variants
17. Product approved → goes live on marketplace
18. Receive order notification → process and dispatch



## 8. API & Integration Requirements

| Integration           | Type                | Requirement                                                             |
|-----------------------|---------------------|-------------------------------------------------------------------------|
| Xerin Logistics API   | REST + Webhooks     | Shipment creation, tracking events, proof of delivery, rate calculation |
| M-Pesa (Safaricom)    | REST API / STK Push | C2B payment initiation, payment confirmation callbacks                  |
| Airtel Money          | REST API            | Payment initiation and status queries across East Africa                |
| MTN MoMo              | REST API            | West/Central Africa mobile money payments                               |
| Flutterwave           | REST API            | Card payments, bank transfers, multi-currency support                   |
| Africa's Talking      | REST API            | SMS OTP delivery, notification SMS, USSD for feature phones             |
| Firebase FCM          | SDK + REST          | Push notification delivery to web and mobile clients                    |
| AWS S3                | SDK                 | Product image upload, storage, and CDN delivery                         |
| Google OAuth          | OAuth 2.0           | Social login for buyers; Google account verification                    |
| Elasticsearch         | REST API            | Product search indexing, full-text queries, and faceted filtering       |
| Currency Exchange API | REST API            | Real-time FX rates for multi-currency display and conversion            |

## 9. Security Requirements

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### 9.1 Authentication & Authorisation

- All API endpoints protected by JWT-based bearer tokens
- Role-Based Access Control (RBAC) enforced at API layer
- Short-lived access tokens (15 min) with refresh token rotation
- 2FA via TOTP (Google Authenticator) or SMS OTP
- Brute-force protection with exponential back-off on login attempts

### 9.2 Data Protection

- All data encrypted in transit via TLS 1.3
- Sensitive data (passwords, payment tokens) encrypted at rest with AES-256
- PII data minimisation — only collect what is necessary
- GDPR-aligned data handling with clear privacy policy and consent management
- Right to erasure: users can request account and data deletion

### 9.3 Payment Security

- PCI-DSS Level 1 compliance — card data never stored on Xerin servers
- Tokenisation via payment gateway provider
- 3D Secure 2.0 for card transactions where supported
- Fraud detection scoring on high-value orders

### 9.4 Application Security

- OWASP Top 10 mitigations implemented and tested
- SQL injection prevention via parameterised queries and ORM
- XSS protection via Content Security Policy headers
- CSRF token validation on all state-changing requests
- Rate limiting on authentication, search, and order endpoints
- Regular third-party penetration testing (minimum annual)
- Dependency vulnerability scanning via Snyk or equivalent

## 10. Future Roadmap

| Phase   | Feature                     | Description                                                                                       | Priority |
|---------|-----------------------------|---------------------------------------------------------------------------------------------------|----------|
| Phase 2 | Xerin Pay                   | Native platform wallet enabling stored-value payments, split payments, and instant seller payouts | High     |
| Phase 2 | Xerin Prime Membership      | Subscription tier offering free shipping, early access to deals, and exclusive discounts          | High     |
| Phase 2 | Android App                 | Native Android application optimised for low-end devices and offline browsing                     | High     |
| Phase 2 | iOS App                     | Native iOS application with Apple Pay and Face ID authentication                                  | Medium   |
| Phase 3 | AI Product Recommendations  | Machine learning engine personalising home feed, search results, and email campaigns              | High     |
| Phase 3 | Vendor Verification System  | Enhanced KYC with biometric verification and business registry API integration                    | Medium   |
| Phase 3 | Warehouse Management System | Fulfilment by Xerin (FBX) — sellers store inventory in Xerin warehouses for faster dispatch       | High     |
| Phase 3 | Affiliate Program           | Referral and influencer marketing platform with tracked links and commission payouts              | Medium   |
| Phase 4 | B2B Marketplace             | Wholesale and bulk-order module for business buyers                                               | Medium   |
| Phase 4 | Live Commerce               | In-app live video shopping with real-time product purchases                                       | Low      |
| Phase 4 | BNPL Expansion              | Buy Now Pay Later integration across all major markets with local partners                        | Medium   |

## 11. Appendix — Glossary

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| Term                     | Definition                                                                                                |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>GMV</b>               | Gross Merchandise Value — total value of goods sold through the platform over a period                    |
| <b>Seller Commission</b> | Percentage of each sale retained by Xerin Market as a platform fee                                        |
| <b>FBX</b>               | Fulfilment by Xerin — Xerin's first-party warehousing and fulfilment service                              |
| <b>STK Push</b>          | Safaricom's M-Pesa API mechanism that sends a payment prompt to a customer's phone                        |
| <b>KYC</b>               | Know Your Customer — identity verification process for sellers                                            |
| <b>PWA</b>               | Progressive Web Application — web app that provides native app-like experience                            |
| <b>SSR</b>               | Server-Side Rendering — pages rendered on the server for faster load and better SEO                       |
| <b>ISR</b>               | Incremental Static Regeneration — Next.js feature for updating static pages on demand                     |
| <b>Webhook</b>           | HTTP callback sent by an external service to notify Xerin of an event (e.g., payment confirmed)           |
| <b>RBAC</b>              | Role-Based Access Control — permission system based on assigned user roles                                |
| <b>PII</b>               | Personally Identifiable Information — data that can identify an individual                                |
| <b>CDN</b>               | Content Delivery Network — distributed network that serves static assets from edge servers close to users |

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— End of Document —

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